Science PRI - Formal Laws and First Weld

Pre-Irreversibility Inhibition as architectural primitive. Named first commercial driver: payout clear.

Source: gate/science_pri.py

Motive

Contribute massively to Tier-S continuity as a favor to global coordinators. Collaborate. Do not replace C2, nukes, grid, or link monopolies.

Formula

> *Inhibit unjustified irreversible acts (PRI) - fail-closed - as a driver-node mouth on the act graph, not a hub dashboard.*

Tech thesis

Frontiers multiply what systems *can* do. Gate owns the mouth between can and may - contribute under each stack; do not own their guns.

First weld

ID: withdraw_payout_clear

Write: withdraw

Label: Withdraw / payout - clear before wire

Why: First commercial path: irreversible money leave. Prove fail-closed clearance on one payout path before grid or programs.

Example path: POST /v1/payouts/{id}/release

Checkout: /operator?write=withdraw

Next after prove: grid_shed_reconnect

Not first:

- grid blackstart (step 2 / Tier-S-adjacent)
- milcomms release mouth (ladder step 4)
- nuclear C2 (state only)

Science claims -> Gate law

Pre-Irreversibility Inhibition (PRI)

Catastrophe often comes from acting when epistemic adequacy cannot justify irreversible commitment. PRI makes restraint an architectural primitive.

Gate law: execute(a) iff LIVE(a) and justified(a); else DENY. bypass(a)=empty. CHARGE-only re-open.

- Pre-Irreversibility Inhibition (PRI) - https://doi.org/10.5281/zenodo.18013606

Fail-closed (fail-secure) under uncertainty

Under control failure, default DENY. Attackers manufacture uncertainty to fish for fail-open. Availability loss beats silent LIVE.

Gate law: uncertainty(u) => DENY. Life-safety overrides are explicit coordinator lanes, not soft-yes.

- Fail-open vs fail-closed authorization - https://authzed.com/blog/fail-open

Thermodynamic cost of causal intervention

Pearl-perfect surgical do(.) is physically impossible. Real intervention is measure+control+dissipation; causation rides the entropy arrow.

Gate law: intervention = measure(evidence) -> control(LIVE|DENY) -> record(receipt). Silent free yes is illegal. Weld + bps price authority dissipation.

- Classical and Quantum Causal Interventions - https://doi.org/10.3390/e20090687

Driver nodes (Liu-Slotine-Barabási)

Network controllability needs a minimum driver set. Drivers often avoid hubs. Sparse heterogeneous systems are hard to control.

Gate law: Gate is a driver on InstitutionalActGraph edges (irreversible writes). Hub dashboards are not drivers. Weld one act-edge at a time.

- Controllability of complex networks - https://doi.org/10.1038/nature10011

Quorum / unfireable mouth

Authority as structured quorum, not one charismatic key. Safety kernels must sit outside the actor's invoke path - fail-closed externally.

Gate law: LIVE <= quorum(policy) AND CHARGE(packet) AND mouth(Gate). Stranger-verify evidence outside actor trust boundary (Velaru).

- Threshold / quorum authorization (distributed custody) - https://arxiv.org/html/2607.08226v1